

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY

(Deemed to be university U/S 3 of UGC Act 1956)

**Accredited with Grade "A++" by NAAC | 12B Status by UGC Approved by AICTE Jeppiaar
Nagar, Rajiv Gandhi Salai, Chennai - 600 119.**



LAB MANUAL

MASTER OF COMPUTER APPLICATIONS

(III SEMESTER)

SBSB6311 BIG DATA ANALYTICS LAB

SATHYABAMA UNIVERSITY

SBSB6311	BIG DATA ANALYTICS LAB	L	T	P	EL	Credits	Total Marks
		0	0	4	0	2	100

List of Experiments

1. Installing and configuring Hadoop on a local or distributed environment - working with HDFS commands
2. Loading data from various sources into Hadoop, such as CSV files, JSON data, or database tables. Algorithms using map reduce - word count, matrix multiplication.
3. Write Pig Latin scripts to process data stored in Hadoop
4. Connecting MongoDB with various database operations and functions.
5. Connect Tableau to a dataset (e.g., CSV file, Excel spreadsheet, and database).
6. Creating interactive dashboards and reports. basic visualizations such as bar charts, line charts, and scatter plots.
7. Customize visualizations by adjusting colors, labels, and formatting options

Program 1: Installing And Configuring Hadoop On A Local Or Distributed Environment - Working With Hdfs Commands

AIM:

To installing and configuring hadoop on a local or distributed environment - working with hdfs commands

ALGORITHM:

- Step 1. **Install Java JDK** on all nodes and set JAVA_HOME.
- Step 2. **Download & extract Hadoop binaries** to a chosen directory.
- Step 3. **Configure hadoop-env.sh** to export JAVA_HOME.
- Step 4. **Edit core-site.xml** to set fs.defaultFS (e.g., hdfs://localhost:9000).
- Step 5. **Edit hdfs-site.xml** to configure dfs.replication and data directories.
- Step 6. **Format the NameNode** using hdfs namenode -format.
- Step 7. **Start HDFS daemons** (start-dfs.sh) and verify with jps.
- Step 8. **Run basic HDFS commands** (hdfs dfs -ls /, -put, -get, -rm).
- Step 9. **Check HDFS web UI** (default http://localhost:9870).
- Step 10. **Scale to cluster** by configuring slaves file and SSH passwordless login.

COMMANDS:

Start HDFS & YARN

Starts NameNode, DataNode, ResourceManager, NodeManager services to run Hadoop cluster. Check with jps to confirm.

Start-all jps

Create a directory in HDFS at given path(s).

Syntax:

hadoop fs -mkdir <paths> Example:

C:\>hadoop fs -mkdir /input_dir

List the contents of a directory.

Syntax :

hadoop fs -ls <args> Example:

C:\>hadoop fs -ls /input_dir/ Upload and

download a file in HDFS.

Upload: **hadoop fs**

-put:

Copy single src file, or multiple src files from local file system to the Hadoop data file system

Syntax:

hadoop fs -put <localsrc> ... <HDFS_dest_Path> **Example:**

C:\>hadoop fs -put E:\input_file.txt /input_dir

Download:

hadoop fs -get:

Copies/Downloads files to the local file system **Syntax:**

hadoop fs -get <hdfs_src> <localdst> **Example:**

C:\>hadoop fs -get /input_dir/input_file.txt /output **Display the aggregate length of a file.**

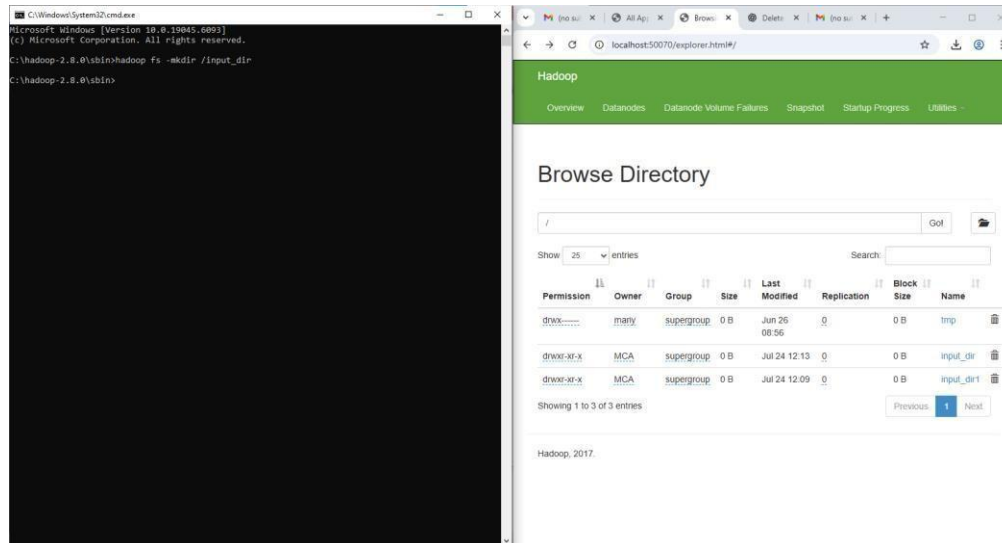
Syntax :

hadoop fs -du <path> **Example:**

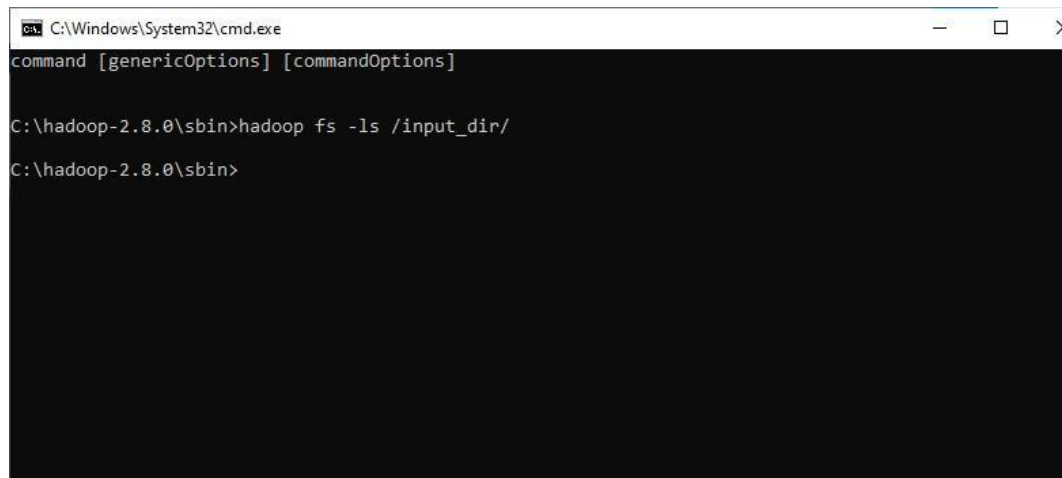
C:\>hadoop fs -du /input_dir/input_file.txt

OUTPUT:

Create a directory in HDFS at given path(s).



List the contents of a directory.



Upload and Download:

```
C:\hadoop-2.8.0\sbin>hadoop fs -put D:\input_file.txt /input_dir
C:\hadoop-2.8.0\sbin>hadoop fs -ls /input_dir/
Found 1 items
-rw-r--r-- 1 MCA supergroup 24 2025-07-24 12:26 /input_dir/input_file.txt
C:\hadoop-2.8.0\sbin>
```

Display the aggregate length of a file.

```
C:\hadoop-2.8.0\sbin>hadoop fs -du /input_dir/input_file.txt
24 /input_dir/input_file.txt
C:\hadoop-2.8.0\sbin>_
```

RESULT:

Thus the program executed successfully and the output was verified.

Program 2: Loading data from various sources into Hadoop, such as CSV files, JSON data, or database tables. Algorithms using map reduce - word count, matrix multiplication.

AIM:

To Loading data from various sources into Hadoop, such as CSV files, JSON data, or database tables. Algorithms using map reduce - word count, matrix multiplication.

ALGORITHM:

- Step 1. **Start Hadoop services**
- Step 2. **Create directory in HDFS**
- Step 3. **Copy local file to HDFS**
- Step 4. **List HDFS directory contents**
- Step 5. **Run MapReduce job (e.g., wordcount)**
- Step 6. **Check output directory in HDFS**
- Step 7. **Display processed output**
- Step 8. **Count total lines (records)**
- Step 9. **Tail last few lines of a file**
- Step 10. **Delete file or directory from HDFS**
- Step 11. **Stop Hadoop services**

COMMANDS:

1.Create a directory in HDFS at given path(s).

Syntax:

hadoop fs -mkdir <paths> Example:

C:\>hadoop fs -mkdir /input_dir

2.Verify: hadoop fs -ls /

Create a file name input.txt in (C:\Users\mcstudent)

3. Upload:

hadoop fs -put C:\Users\mcstudent\input.txt /input_dir/

check the directory in : <http://localhost:50070/>

4.List the contents of a directory.

Syntax :

hadoop fs -ls <args> Example:

C:\>hadoop fs -ls /input_dir/ Upload and download a file in HDFS.

Upload: hadoop fs

-put:

5. Copy single src file, or multiple src files from local file system to the Hadoop data file system

Syntax: hadoop fs -put <localsrc> ... <HDFS_dest_Path>

Example:

C:\>hadoop fs -put E:\input_file.txt /input_dir

6.Word count

A common MapReduce program that counts words in input files

Syntax: hadoop jar <path_to_examples_jar> wordcount <input_path> <output_path>

Example:

hadoop jar /path/hadoop-mapreduce-examples.jar wordcount /input /output **Total records**

7.Count lines (records) in a file.

Syntax:. hdfs dfs -cat
<hdfs_file_path>wc -l

Example:

hdfs dfs -cat /user/hadoop/file.txt | wc -l

8.Tail data

Show last few lines of a file in HDFS.

Syntax:. hdfs dfs -tail
<hdfs_file_path>

Example:

hdfs dfs -tail /path/file.txt

9.Delete file

Remove files or directories from HDFS.

Syntax:. hdfs dfs -rm

<hdfs_file_path>

Example:

```
hdfs dfs -rm /path/file.txt
```

10. Length:

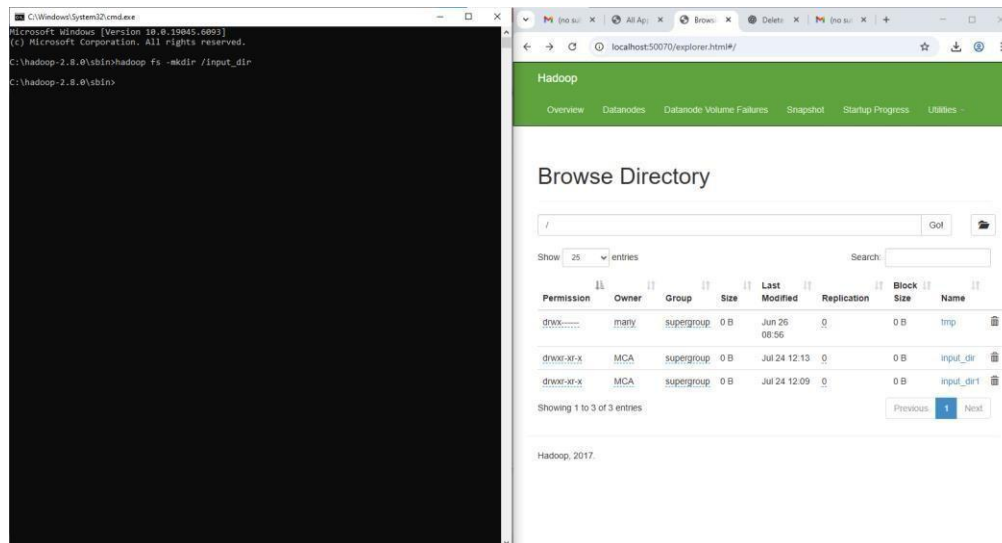
```
hadoop fs -du -h /input_dir/input.txt
```

11. Remove file:

```
hadoop fs -rm /input_dir/new_input.txt
```

OUTPUT:

Create a directory in HDFS at given path(s).



The screenshot shows a Windows command prompt window on the left and a Hadoop web interface on the right. The command prompt shows the execution of the command `hadoop fs -mkdir /input_dir` in the `C:\hadoop-2.8.0\bin` directory. The Hadoop web interface, titled "Browse Directory", shows the contents of the `/` directory. It lists three entries: `drwxr-xr-x` `mary` `supergroup` `0 B` `Jun 26 08:56` `0` `0 B` `tmp`, `drwxr-xr-x` `MCA` `supergroup` `0 B` `Jul 24 12:13` `0` `0 B` `input_dir`, and `drwxr-xr-x` `MCA` `supergroup` `0 B` `Jul 24 12:09` `0` `0 B` `input_dir1`. The interface also shows "Showing 1 to 3 of 3 entries" and "Hadoop, 2017."

List the contents of a directory.

```
C:\Windows\System32\cmd.exe
command [genericOptions] [commandOptions]

C:\hadoop-2.8.0\sbin>hadoop fs -ls /input_dir/

C:\hadoop-2.8.0\sbin>
```

Upload

```
C:\Windows\System32\cmd.exe

C:\hadoop-2.8.0\sbin>hadoop fs -put C:\Users\MCA.L2-MBA-12\Downloads\Students.csv /input_dir

C:\hadoop-2.8.0\sbin>s
```

Tail:

```
C:\hadoop-2.8.0\sbin>hadoop fs -tail /input_dir/Students.csv
Undergraduate,Denmark,4.6,Twitter,No,7.3,7,In Relationship,2,5
692,24,Male,Graduate,Ireland,5.9,Instagram,Yes,6.5,6,Single,3,7
693,19,Female,Undergraduate,India,7.0,TikTok,Yes,5.8,4,Single,5,9
694,22,Male,Graduate,Switzerland,4.5,Facebook,No,7.3,7,In Relationship,2,5
695,20,Female,Undergraduate,Turkey,6.6,Instagram,Yes,6.1,5,Single,4,8
696,23,Male,Graduate,USA,5.5,Twitter,Yes,6.7,6,In Relationship,3,7
697,21,Female,Undergraduate,Mexico,6.3,TikTok,Yes,6.2,5,Single,4,8
698,24,Male,Graduate,France,4.8,Facebook,No,7.1,7,In Relationship,2,5
699,19,Female,Undergraduate,Canada,5.7,Instagram,Yes,6.6,6,Single,3,7
700,22,Male,Graduate,UK,6.2,Twitter,Yes,6.3,5,Single,4,8
701,20,Female,Undergraduate,Italy,4.7,TikTok,No,7.2,7,In Relationship,2,5
702,23,Male,Graduate,Russia,6.8,Instagram,Yes,5.9,4,Single,5,9
703,21,Female,Undergraduate,China,5.6,WeChat,Yes,6.7,6,In Relationship,3,7
704,24,Male,Graduate,Japan,4.3,Twitter,No,7.5,8,Single,2,4
705,19,Female,Undergraduate,Poland,6.2,Facebook,Yes,6.3,5,Single,4,8
```

Total records:

```
C:\hadoop-2.8.0\sbin>hadoop fs -put C:\Users\MCA.L2-MBA-12\Downloads\Students.csv /input_dir

C:\hadoop-2.8.0\sbin>hadoop fs -cat /input_dir/Students.csv
Student_ID,Age,Gender,Academic_Level,Country,Avg_Daily_Usage_Hours,Most_Used_Platform,Affects_Academic_Performance,Sleep_Hours_Per_Night,Mental_Health_Score,Relationship_Status,Conflicts_Over_Social_Media,Addicted_Score
1,19,Female,Undergraduate,Bangladesh,5.2,Instagram,Yes,6.5,6,In_Relationship,3,8
2,22,Male,Graduate,India,2.1,Twitter,No,7.5,8,Single,0,3
3,20,Female,Undergraduate,USA,6.0,TikTok,Yes,5.0,5,Complicated,4,9
4,18,Male,High_School,UK,3.0,YouTube,No,7.0,7,Single,1,4
5,21,Male,Graduate,Canada,4.5,Facebook,Yes,6.0,6,In_Relationship,2,7
6,19,Female,Undergraduate,Australia,7.2,Instagram,Yes,4.5,4,Complicated,5,9
7,23,Male,Graduate,Germany,1.5,LinkedIn,No,8.0,9,Single,0,2
8,20,Female,Undergraduate,Brazil,5.8,Snapchat,Yes,6.0,6,In_Relationship,2,8
9,18,Male,High_School,Japan,4.0,TikTok,No,6.5,7,Single,1,5
10,21,Female,Graduate,South_Korea,3.3,Instagram,No,7.0,7,In_Relationship,1,4
11,19,Male,Undergraduate,France,4.8,Snapchat,Yes,6.2,5,Complicated,3,7
12,20,Female,Undergraduate,Spain,5.5,TikTok,Yes,5.8,6,In_Relationship,2,8
13,22,Male,Graduate,Italy,2.8,LinkedIn,No,7.2,8,Single,1,4
14,18,Female,High_School,Mexico,6.5,Instagram,Yes,5.5,5,Single,4,9
15,21,Male,Undergraduate,Russia,3.7,YouTube,No,6.8,7,In_Relationship,2,5
16,20,Female,Undergraduate,China,4.2,TikTok,Yes,6.0,6,Complicated,3,7
17,24,Male,Graduate,Sweden,2.0,LinkedIn,No,7.8,8,Single,0,3
18,19,Female,High_School,Norway,5.0,Instagram,Yes,5.7,5,In_Relationship,3,8
19,21,Male,Undergraduate,Denmark,3.5,Facebook,No,6.7,7,Single,1,5
20,20,Female,Undergraduate,Netherlands,4.7,Snapchat,Yes,5.9,6,Complicated,3,7
21,18,Male,High_School,Belgium,5.3,TikTok,Yes,5.5,5,Single,4,8
22,23,Female,Graduate,Switzerland,2.5,LinkedIn,No,7.3,8,In_Relationship,1,4
23,19,Male,Undergraduate,Austria,4.9,Instagram,Yes,5.8,6,Complicated,3,7
24,20,Female,Undergraduate,Portugal,5.7,TikTok,Yes,5.4,5,Single,4,8
25,22,Male,Graduate,Greece,3.2,Facebook,No,6.9,7,In_Relationship,2,5
26,19,Female,High_School,Ireland,6.1,Instagram,Yes,5.2,5,Complicated,4,9
27,21,Male,Undergraduate,New_Zealand,3.8,YouTube,No,6.6,7,Single,1,5
28,20,Female,Undergraduate,Singapore,4.4,TikTok,Yes,5.9,6,In_Relationship,3,7
29,24,Male,Graduate,Malaysia,2.2,LinkedIn,No,7.4,8,Single,0,3
30,19,Female,High_School,Thailand,5.9,Instagram,Yes,5.3,5,Complicated,4,8
31,21,Male,Undergraduate,Vietnam,3.6,Facebook,No,6.7,7,Single,1,5
32,20,Female,Undergraduate,Philippines,4.8,Snapchat,Yes,5.7,6,In_Relationship,3,7
33,18,Male,High_School,Indonesia,5.4,TikTok,Yes,5.4,5,Complicated,4,8
34,23,Female,Graduate,Taiwan,2.6,LinkedIn,No,7.2,8,Single,1,4
35,19,Male,Undergraduate,Hong_Kong,4.7,Instagram,Yes,5.8,6,In_Relationship,3,7
36,20,Female,Undergraduate,Turkey,5.6,TikTok,Yes,5.5,5,Complicated,4,8
37,22,Male,Graduate,Israel,3.1,Facebook,No,6.8,7,Single,1,5
```

Delete:

```
C:\hadoop-2.8.0\sbin>hadoop fs -rm /input_dir/Students.csv
Deleted /input_dir/Students.csv
```

```
C:\hadoop-2.8.0\sbin>
```

RESULT:

Thus the program executed successfully and the output was verified.

Program 3: Write Pig Latin Scripts to Process Data Stored in Hadoop

AIM:

To write and execute Pig Latin scripts that process data stored in the Hadoop Distributed File System (HDFS).

ALGORITHM:

- Step 1. **Start** the Hadoop and Pig services.
- Step 2. **Upload** the input data file to HDFS.
- Step 3. **Open** the Pig Grunt shell or run in script mode.
- Step 4. **Load** the HDFS data into a Pig relation using the `LOAD` statement.

Step 5. **Perform** the desired transformation or computation using Pig Latin commands (`FILTER`, `GROUP`, `FOREACH`, `ORDER`).

Step 6. **Store or Dump** the result for viewing or saving into HDFS. Step 7. **Stop** the Pig shell and view the result.

PROCEDURE:

Let us assume a dataset: `students.txt`

Contents of students.txt (stored in local/HDFS):

```
StudentID,Name,Marks
```

```
101,Ravi,85
```

```
102,Anita,92
```

```
103,Suresh,58
```

```
104,Kavya,76
```

```
105,Manoj,45
```

Step 1: Upload file to HDFS

```
hdfs dfs -mkdir /pigexample
```

```
hdfs dfs -put students.txt /pigexample/
```

Step 2: Open Pig (in local or mapreduce mode)

```
Pig
```

Step 3: Pig Latin Script --

Load the file from HDFS

```
students = LOAD '/pigexample/students.txt' USING PigStorage(',')
          AS (id:int, name:chararray, marks:int);
```

```
-- Filter students who scored more than 60
passed_students = FILTER students
                  BY marks > 60;
```

```
-- Order students by marks descending
ordered_students = ORDER passed_students
                   BY marks DESC;
```

```
-- Display the result
DUMP ordered_students;
```

Step 4: Run the Script in Batch Mode pig student_filter.pig

OUTPUT:

(102,Anita,92)
(101,Ravi,85)
(104,Kavya,76)

RESULT:

The Pig Latin script was successfully written and executed to filter and sort student data stored in HDFS. The result displayed students with marks greater than 60, ordered in descending order.

Program 4: Connecting MONGODB Database with various database operations and functions.

AIM:

To Connecting MONGODB Database with various database operations and functions.

ALGORITHM:

Step1. Download a dataset from a trusted source or create one in Excel.

Step2. Save the dataset as .csv or .xlsx format.

Step3. Set up your SQL database (e.g., MySQL, PostgreSQL).

Step4. Create a new database using CREATE DATABASE in SQL.

Step5. Import the dataset using LOAD DATA Import

Step6. Perform operations like SELECT, INSERT, WHERE, ORDER BY, or GROUP BY to manipulate data.

Step7. Use SQL functions like COUNT(),CONCAT(),MAX(),MIN(),TRIM() for transformations. Step8. Export results with SELECT INTO OUTFILE if needed.

PROCEDURE:

Open your SQL client or use a terminal for the DBMS.

Create a new database:

```
CREATE DATABASE MyDatabase;
```

```
USE MyDatabase;
```

Import the dataset: a.

If .csv:

```
LOAD DATA INFILE 'path_to_file.csv'
INTO TABLE EmployeeData
FIELDS TERMINATED BY ',' ENCLOSED
BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```

b. If .xlsx:

i. Convert it to .csv or use a library/tool like pandas in Python to load it into the database.

SQL Operations Retrieve Data:

```
SELECT * FROM EmployeeData;
```

Filter Data:

```
SELECT * FROM EmployeeData WHERE Department = 'IT';
```

Sort Data:

```
SELECT * FROM EmployeeData ORDER BY Salary DESC;
```

Group and Aggregate Data:

```
SELECT Department, AVG(Salary) AS AvgSalary FROM EmployeeData
GROUP BY Department;
```

Insert:

```
INSERT INTO EmployeeData (EmployeeID, Name, Department, Age, JoiningDate, Salary)
VALUES (31, 'Rohan Nair', 'IT', 28, '2023-01-10', 48000);
```

Use Functions:

COUNT()

```
SELECT COUNT(*) AS TotalEmployees FROM EmployeeData;
```

CONCAT()

```
SELECT CONCAT(Name, ' (', Department, ')') AS EmployeeInfo FROM
EmployeeData;
```

MAX()

```
SELECT MAX(Salary) AS HighestSalary FROM EmployeeData;
```

MIN()

```
SELECT MIN(Salary) AS LowestSalary FROM EmployeeData;
```

TRIM()

```
SELECT TRIM(Department) AS CleanedDepartment FROM
EmployeeData
```

OUTPUT:

EmployeeID	Name	Department	Age	JoiningDate	Salary
1	Ananya Sharma	HR	28	2021-03-15	45000.00
2	Ravi Kumar	IT	32	2019-06-10	65000.00
3	Sneha Joshi	Finance	30	2020-09-25	50000.00
4	Arjun Mehta	IT	25	2022-01-12	40000.00
5	Pooja Reddy	HR	29	2020-05-18	46000.00
6	Vivek Anand	Marketing	33	2018-08-22	55000.00
7	Priya Singh	Finance	27	2021-11-03	48000.00
8	Karan Gupta	IT	31	2019-03-20	62000.00
9	Neha Patil	HR	26	2022-04-10	42000.00
10	Aditya Rao	Marketing	34	2017-10-05	58000.00
11	Riya Kapoor	Finance	28	2021-02-17	50000.00
12	Mohit Verma	IT	29	2018-11-23	60000.00
13	Kavita Jain	HR	35	2016-07-29	47000.00
14	Rahul Desai	Marketing	30	2020-12-11	54000.00
15	Shweta Bansal	Finance	32	2019-09-30	51000.00
16	Rajeev Nair	IT	27	2021-06-15	45000.00
17	Meera Krishnan	HR	28	2022-02-20	43000.00
18	Anil Kumar	Marketing	34	2017-12-10	56000.00
19	Sunita Yadav	Finance	29	2020-07-17	49000.00
20	Rohit Sharma	IT	33	2018-03-14	66000.00
21	Anuj Thakur	Marketing	35	2016-11-23	57000.00
22	Neeraj Gupta	HR	30	2021-08-29	44000.00
23	Radhika Iyer	Finance	31	2019-01-07	52000.00
24	Sanjay Chawla	IT	26	2022-03-21	47000.00
25	Tanya Arora	Marketing	28	2021-04-11	53000.00

EmployeeID	Name	Department	Age	JoiningDate	Salary
2	Ravi Kumar	IT	32	2019-06-10	65000.00
4	Arjun Mehta	IT	25	2022-01-12	40000.00
8	Karan Gupta	IT	31	2019-03-20	62000.00
12	Mohit Verma	IT	29	2018-11-23	60000.00
16	Rajeev Nair	IT	27	2021-06-15	45000.00
20	Rohit Sharma	IT	33	2018-03-14	66000.00
24	Sanjay Chawla	IT	26	2022-03-21	47000.00
28	Pankaj Tiwari	IT	29	2018-09-30	63000.00

EmployeeID	Name	Department	Age	JoiningDate	Salary
20	Rohit Sharma	IT	33	2018-03-14	66000.00
2	Ravi Kumar	IT	32	2019-06-10	65000.00
28	Pankaj Tiwari	IT	29	2018-09-30	63000.00
8	Karan Gupta	IT	31	2019-03-20	62000.00
12	Mohit Verma	IT	29	2018-11-23	60000.00
10	Aditya Rao	Marketing	34	2017-10-05	58000.00
21	Anuj Thakur	Marketing	35	2016-11-23	57000.00
18	Anil Kumar	Marketing	34	2017-12-10	56000.00
27	Manish Patel	Finance	33	2017-06-16	55000.00
6	Vivek Anand	Marketing	33	2018-08-22	55000.00
14	Rahul Desai	Marketing	30	2020-12-11	54000.00
29	Kiran Sahu	Marketing	30	2020-01-25	54000.00
25	Tanya Arora	Marketing	28	2021-04-11	53000.00
23	Radhika Iyer	Finance	31	2019-01-07	52000.00
15	Shweta Bansal	Finance	32	2019-09-30	51000.00
11	Riya Kapoor	Finance	28	2021-02-17	50000.00
3	Sneha Joshi	Finance	30	2020-09-25	50000.00
19	Sunita Yadav	Finance	29	2020-07-17	49000.00
7	Priya Singh	Finance	27	2021-11-03	48000.00
24	Sanjay Chawla	IT	26	2022-03-21	47000.00
13	Kavita Jain	HR	35	2016-07-29	47000.00
5	Pooja Reddy	HR	29	2020-05-18	46000.00
16	Rajeev Nair	IT	27	2021-06-15	45000.00
1	Ananya Sharma	HR	28	2021-03-15	45000.00
22	Neeraj Gupta	HR	30	2021-08-29	44000.00

Department	AvgSalary
Finance	50714.285714
HR	43750.000000
IT	56000.000000
Marketing	55285.714286

✓ 1 row inserted. (Query took 0.0007 seconds.)

INSERT INTO EmployeeData (EmployeeID, Name, Department, Age, JoiningDate, Salary) **VALUES** (31, 'Rohan Nair', 'IT', 28, '2023-01-10', 48000);

TotalEmployees

31

EmployeeInfo

Ananya Sharma (HR)

Ravi Kumar (IT)

Sneha Joshi (Finance)

Arjun Mehta (IT)

Pooja Reddy (HR)

Vivek Anand (Marketing)

Priya Singh (Finance)

Karan Gupta (IT)

Neha Patil (HR)

Aditya Rao (Marketing)

Riya Kapoor (Finance)

Mohit Verma (IT)

Kavita Jain (HR)

Rahul Desai (Marketing)

Shweta Bansal (Finance)

Rajeev Nair (IT)

Meera Krishnan (HR)

Anil Kumar (Marketing)

Sunita Yadav (Finance)

Rohit Sharma (IT)

Anuj Thakur (Marketing)

Neeraj Gupta (HR)

Radhika Iyer (Finance)

Sanjay Chawla (IT)

Tanya Arora (Marketing)

HighestSalary

66000.00

CleanedDepartment

HR

IT

Finance

IT

HR

Marketing

Finance

IT

HR

Marketing

Finance

IT

HR

Marketing

Finance

IT

HR

Marketing

Finance

IT

Marketing

HR

Finance

IT

Marketing

RESULT:

Thus the above program executed successfully and output verified.

Program 5: Connecting Tableau to a Dataset (CSV, Excel Sheet, or Database)

AIM:

To establish a connection between Tableau and various data sources such as CSV files, Excel spreadsheets, or databases, enabling data visualization and analysis.

ALGORITHM:

1. **Start Tableau Desktop.**
2. **Choose a Data Source:**
 - For CSV/Excel: Select “**File**” and choose the file from the system.
 - For Database: Select “**Server**”, choose the appropriate database type (e.g., MySQL, PostgreSQL), and enter credentials.
3. **Connect to Data:** ○ Tableau imports the selected dataset into the **Data Source** tab.
4. **Inspect the Schema:**
 - Preview data to check column names, types, and values.
5. **Drag required tables (if needed) into canvas.**
6. **Click “Sheet” to proceed to visualization.**
7. **Save Workbook for future use.**

PROCEDURE:

For CSV or Excel Sheet:

1. Open Tableau Desktop.
2. On the **Start Page**, under **Connect**, click **Microsoft Excel** or **Text File (CSV)**.
3. Navigate to and select the desired file (e.g., sales_data.xlsx or data.csv).
4. Tableau loads the file into the **Data Source** tab.
5. You can view the data in tabular format and make any adjustments (rename columns, data types, etc.).
6. Click on **Sheet 1** at the bottom to create visualizations using the connected data.

For Database (e.g., MySQL/PostgreSQL):

1. Open Tableau Desktop.
2. Under **Connect**, click the appropriate database type (e.g., **MySQL**).
3. Enter the following: ○ **Server name/IP address** ○ **Port number** ○

Username and password ○ **Database name**

4. Click **Sign In**.
5. Once connected, select the desired table(s) to load.
6. Drag tables into the join area if multiple tables are needed.
7. Click on **Sheet 1** to start building visualizations.

OUTPUT:

The screenshot shows the Tableau Desktop Public Edition interface. The top menu bar includes 'File', 'Data', 'Window', and 'Help'. The main window title is 'Tableau Desktop Public Edition'. On the left sidebar, the 'Connections' section shows 'student' as a text file connection. The 'Files' section lists 'data1.csv', 'input_file.txt.txt', and 'student.csv'. The main area displays a table named 'student' with 13 fields and 705 rows. The table columns are: Student ID, Age, Gender, Academic Level, Country, Avg Daily Usage Hours, Most Used Platform, Affects Academic Perform, Sleep Hours Per Night, and Mer. The table data is as follows:

Student ID	Age	Gender	Academic Level	Country	Avg Daily Usage Hours	Most Used Platform	Affects Academic Perform	Sleep Hours Per Night	Mer
1	19	Female	Undergraduate	Bangladesh	5.20000	Instagram	Yes	6.50000	
2	22	Male	Graduate	India	2.10000	Twitter	No	7.50000	
3	20	Female	Undergraduate	USA	6.00000	TikTok	Yes	5.00000	
4	18	Male	High School	UK	3.00000	YouTube	No	7.00000	
5	21	Male	Graduate	Canada	4.50000	Facebook	Yes	6.00000	
6	19	Female	Undergraduate	Australia	7.20000	Instagram	Yes	4.50000	
7	23	Male	Graduate	Germany	1.50000	LinkedIn	No	8.00000	
8	20	Female	Undergraduate	Brazil	5.80000	Snapchat	Yes	6.00000	
9	18	Male	High School	Japan	4.00000	TikTok	No	6.50000	
10	21	Female	Graduate	South Korea	3.30000	Instagram	No	7.00000	
11	19	Male	Undergraduate	France	4.80000	Snapchat	Yes	6.20000	
12	20	Female	Undergraduate	Spain	5.50000	TikTok	Yes	5.80000	
13	22	Male	Graduate	Italy	2.80000	LinkedIn	No	7.20000	
14	18	Female	High School	Mexico	6.50000	Instagram	Yes	5.50000	
15	21	Male	Undergraduate	Russia	3.70000	YouTube	No	6.80000	
16	20	Female	Undergraduate	China	4.20000	TikTok	Yes	6.00000	
17	24	Male	Graduate	Sweden	2.00000	LinkedIn	No	7.80000	

RESULT:

Tableau successfully connected to the selected dataset (CSV/Excel/Database). The data was imported into Tableau's workspace, and visualizations were created based on the connected data source.

Program 6: Creating interactive dashboards and reports. basic visualizations such as bar charts, line charts, and scatter plots.

AIM

To create interactive dashboards and analytical reports in Tableau using basic visualizations such as bar charts, line charts, and scatter plots.

ALGORITHM

1. **Start Tableau Desktop.**
 2. **Connect to a Data Source (CSV, Excel, or Database).**
 3. **Load the dataset and navigate to a new worksheet.**
 4. **Drag and drop dimensions and measures to build:**
 - A **Bar Chart** for categorical comparison
 - A **Line Chart** for trend over time
 - A **Scatter Plot** for correlation analysis
 5. **Rename and organize worksheets.**
 6. **Create a new Dashboard.**
 7. **Drag visualizations onto the dashboard.**
 8. **Add filters or interactivity if needed.**
 9. **Save and publish the dashboard or export the report.**
-

PROCEDURE

Step 1: Connect to Data

1. Open Tableau Desktop.
2. Connect to a dataset (e.g., sales_data.csv or superstore.xlsx).

3. Load the required table(s) into the data pane.

Step 2: Create Basic Visualizations

A. Bar Chart

1. Go to **Worksheet 1**.
2. Drag a **Dimension** (e.g., Category) to the **Columns** shelf.
3. Drag a **Measure** (e.g., Sales) to the **Rows** shelf.
4. Tableau will automatically generate a bar chart.
5. Rename the sheet to “Bar Chart”.

B. Line Chart

1. Open **Worksheet 2**.
2. Drag a **Date field** (e.g., Order Date) to the **Columns** shelf.
3. Drag a **Measure** (e.g., Profit) to the **Rows** shelf.
4. Click the “**Show Me**” panel and select the **Line Chart**.
5. Rename the sheet to “Line Chart”.

C. Scatter Plot

1. Open **Worksheet 3**.
2. Drag one **Measure** (e.g., Sales) to the **Columns** shelf.
3. Drag another **Measure** (e.g., Profit) to the **Rows** shelf.
4. Drag a **Dimension** (e.g., Region) to the **Color** mark to distinguish groups.
5. Rename the sheet to “Scatter Plot”.

Step 3: Create Dashboard

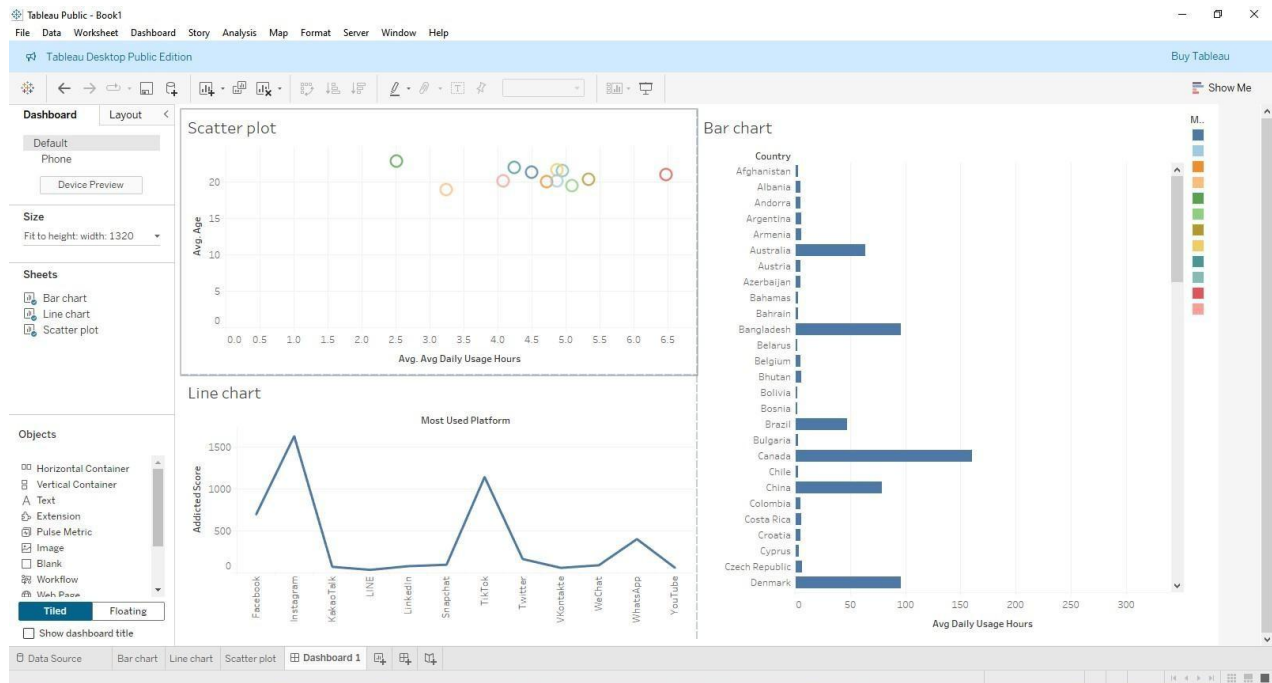
1. Click the **Dashboard** tab and choose “**New Dashboard**”.
2. Drag the “Bar Chart”, “Line Chart”, and “Scatter Plot” worksheets onto the dashboard.

3. Resize and arrange the charts for clarity.
4. Add **Filters**, **Legends**, or **Titles** as needed.
5. Add interactivity like **filter actions** or **highlight actions**.

Step 4: Save and Export

1. Save the Tableau Workbook (.twb or .twbx).
2. Export the dashboard as PDF/image or publish to Tableau Public/Server if required.

OUTPUT:



RESULT:

Successfully created and combined basic visualizations (Bar Chart, Line Chart, and Scatter Plot) into an interactive Tableau dashboard. The dashboard allows users to explore data trends, compare categories, and analyze correlations visually.

Program 7: Customize visualizations by adjusting colors, labels, and formatting options.

Aim:

To Customize visualizations by adjusting colors, labels, and formatting options.

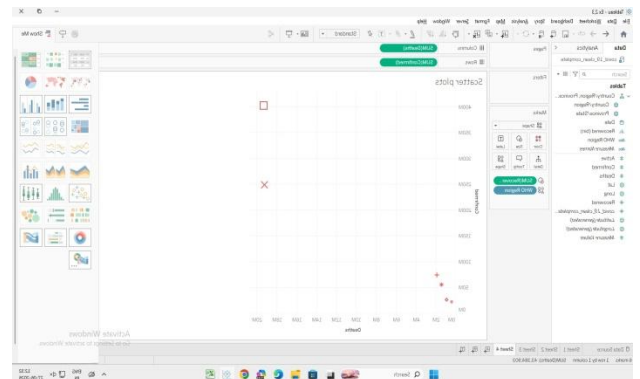
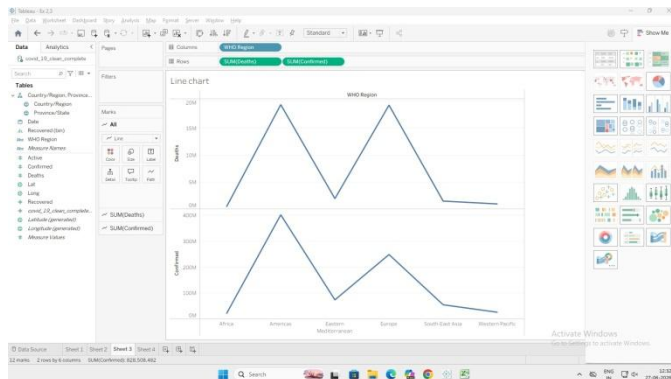
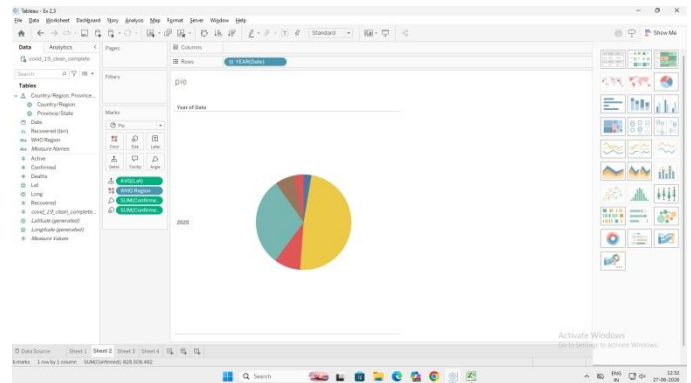
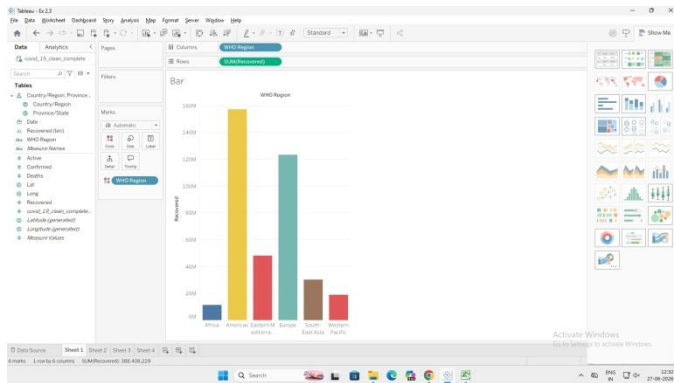
Algorithm:

1. Start Tableau Desktop.
 2. Connect to a Data Source (CSV, Excel, or Database).
 3. Load the dataset and navigate to a new worksheet.
 4. Drag and drop dimensions and measures to build
 5. Create a new Dashboard.
 6. Drag visualizations onto the dashboard.
 7. Add filters or interactivity if needed.
 8. Save and publish the dashboard or export the report.
-

PROCEDURE:

1. Click the Dashboard tab and choose “New Dashboard”.
2. Drag the “Bar Chart”, “Line Chart”, and “Scatter Plot” worksheets onto the dashboard.
3. Resize and arrange the charts for clarity.
4. Add Filters, Legends, or Titles as needed.
5. Use the needed colors, filters, labels and formatting options
6. Add interactivity like filter actions or highlight actions.

OUTPUT:



RESULT:

Thus the program executed successfully and the output was verified.